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Question Paper Code : 20629

B.E. / B.TECH. DEGREE EXAMINATIONS, NOVEMBER / DECEMBER 2023

Third Semester

Computer Science and Engineering

19CS303 – COMPUTER ORGANIZATION AND ARCHITECTURE

(Regulations: Mepco – R2019)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

BTL, CO

PART A – (10 × 2 = 20 Marks)

A, CO1 1. Find the 2's complement of +120.

A,CO1 2. Consider a hypothetical processor with an instruction of type LW R1, 20(R2), which during execution reads a 32-bit word from memory and stores it in a 32-bit register R1. The effective address of the memory location is obtained by the addition of a constant 20 and the contents of register R2. Which of the following best reflects the addressing mode implemented by this instruction for operand in memory? Justify your answer.

A) Indexed addressing B) Register addressing

C) Register indirect scaled addressing D) Base indexed addressing

R,CO2 3. Carry save addition is used for performing _____ operation. Justify your answer.

U,CO2 4. Convert the given decimal number $(19)_{10}$ to binary.

U,CO3 5. Represent the following in Register Transfer notation:

A) The contents of Program counter incremented by 4

B) Transfer the contents of contents from Register R1 to R2

U,CO3 6. Performance of a pipelined processor suffers if _____. Justify your answer.

A) The pipeline stages have different delays

B) Consecutive instructions are dependent on each other

C) The pipeline stages share hardware resources

D) All of the above

A,CO4 7. How many 32K x 1 RAM chips are needed to provide a memory capacity of 256K-bytes? Justify your answer.

A) 8 B) 32 C) 64 D) 128

A,CO4 8. For inclusion to hold between two cache levels L1 and L2 in a multi-level cache hierarchy, which of the following are necessary? Justify your answer.

I. L1 must be a write-through cache

II. L2 must be a write-through cache

III. The associativity of L2 must be greater than that of L1

IV. The L2 cache must be at least as large as the L1 cache

A) IV only B) I and IV only

C) I, II and III only D) I, III and IV only

U,CO5 9. In what way USB differs from that of HDMI interface?

U,CO5 10. What is the significance of edge-triggered interrupt?

PART B – (5 × 16 = 80 Marks)

U, CO1 11. a) i. Explain the following with a necessary example and diagram:

A) Basic operational concept of computer.

B) Functional units of computer. (8 Marks)

A, CO1 11. a) ii. To execute the machine instruction Add LOCA, R0 what are the basic steps needed? Illustrate the same. (8 Marks)

OR

U, CO1 11. b) i. Explain in detail the various instruction formats with an example. (8 Marks)

A, CO1 11. b) ii. Calculate effective address for the following instruction:

A) LOAD 20(R1),R5

B) MOVE #3000,R5

C) Store R5,30(R1,R2)

D) Add –(R2),R5

E) SUB (R1)+,R5 where R1 = 1200 and R2 = 4600 (8 Marks)

A,CO2 12. a) Can you construct a 16-bit carry look ahead adder from 4-bit adders with generate and propagate functions? If so, explain the same. (16 Marks)

OR

A,CO2 12. b) Write the restoring and non-restoring division algorithms. Show how to perform the non – restoring division for 5-bit unsigned integers: 1010/0010. (16 Marks)

A,CO3 13. a) i. Write the control sequence for the instruction

Move(R1), R2

which performs the task of fetching a word from memory? (8 Marks)

U,CO3 13. a) ii. Draw and explain the multiple bus organization while executing the machine instruction MUL R2, R3, R4. (8 Marks)

OR

A,CO3 13. b) i.

MUL R2, R3, R4
ADD R5, R4, R6

Illustrate the 4-stage pipeline execution process of the above instruction sequence. What effect will happen in pipelining process? Apply the suitable technique to resolve the situation. (8 Marks)

U,CO3 13. b) ii. Consider a processor that uses the 4-state branch prediction mechanism. The initial state is either LT or LNT, depending on information provided in the branch instruction. Discuss how the compiler should handle the branch instructions used to control "do while" and "do until" loops and discuss the suitability of the branch prediction mechanism in each case. (8 Marks)

U,CO4 14. a) Discuss the internal organization of semiconductor memory chips with an appropriate diagram. (16 Marks)

OR

U,CO4 14. b) Is it true that cache memory replacement algorithms adopt the principle of locality of reference? If so, how? (16 Marks)

U,CO5 15. a) i. Sketch and explain the contents of registers involved in Direct Memory Access. (8 Marks)

U,CO5 15. a) ii. Discuss the SCSI and PCI bus interface standards. (8 Marks)

OR

U,CO5 15. b) i. What is program controlled I/O? How does it differ from memory mapped I/O? (8 Marks)

U,CO5 15. b) ii. Explain how the interrupt request is handled by the device? (8 Marks)